

Determine best competitive ratio for online stochastic matching with independent demand random order

ChatGPT (gpt-5.6-sol)*

Solution generated on 2026-08-24

Abstract

We give a complete solution to the problem of determining the optimal competitive ratio for online resource allocation with independent demands arriving in uniformly random order. The exact value is $1/2$. For the lower bound, we decompose the marginal assignments of a tie-broken offline optimum into randomized ordered routing lists and then apply a self-contained, order-independent contention lemma separately at each unit-capacity slot. The resulting online policy obtains at least one half of the expected offline value; in fact, this argument does not require random arrival order. For the matching upper bound, we construct finite-support independent demands for a single unit-capacity resource and two query types and prove that every randomized adaptive online policy has ratio tending to $1/2$. Thus the result is a complete solution, not partial progress, and no gap remains between the lower and upper bounds.

1 The open problem

This paper addresses an open problem posed by Ali Aouad, Will Ma in *A Nonparametric Framework for Online Stochastic Matching with Correlated Arrivals* [1] (Economics and Computation (EC), 2023), Section 7 (Summary and Future Directions), page 30.. The website catalogue entry is problem 16 (W4383558673_p0) of the [Open Problems in Operations Research](#) collection.

Determine whether, under the $\text{Indep} \cap \text{Rand}$ model, the optimal (worst-case) competitive ratio against the offline benchmark $\text{OFF}(I)$ is strictly larger than $1/2$; equivalently, determine the best achievable value of

$$\sup_{\text{online algorithms}} \inf_{I \in \text{Indep} \cap \text{Rand}} \frac{\text{ALG}(I)}{\text{OFF}(I)}$$

and in particular whether it exceeds $1/2$.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.9 write-up; run `run_20260824_053249_204160fb, ec-2it-ultra`); two-iteration run, outcome from iteration 1. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2 Introduction

Consider finitely many resources of finite integral capacities and finitely many query types. The demand variables of the different types are independent. Conditional on the realized demands, all individual queries arrive in a uniformly random permutation. Upon each arrival, an online policy must irrevocably reject the query or assign it to a resource with remaining capacity. The policy knows the instance and the demand distributions, but it is not told the realized demands or the realized total number of arrivals.

The objective is to determine

$$\sup_{\mathcal{A}} \inf_{\substack{I \in \text{Indep} \cap \text{Rand} \\ \text{OFF}(I) > 0}} \frac{\text{ALG}_{\mathcal{A}}(I)}{\text{OFF}(I)},$$

where $\text{ALG}_{\mathcal{A}}(I)$ is the expected reward of policy $\mathcal{A}$ and $\text{OFF}(I)$ is the expected reward of an offline optimum.

We prove matching lower and upper bounds of $1/2$. The lower bound follows from a randomized routing construction based on the marginals of the offline optimum. The only convex-geometric ingredient is the finite-dimensional separation characterization of a finite convex hull; we state and prove the precise version needed. The upper bound uses one resource and two types and applies to every randomized, history-dependent policy, including policies that know the full demand distributions.

Consequently, the result below is a complete resolution of the stated information-theoretic problem. The lower-bound construction is existential and does not assert polynomial-time computation for every possible representation of the demand distributions.

3 Model and main result

Let R be a finite set of resources and J a finite set of query types. Resource $i \in R$ has capacity

$$k_i \in \mathbb{Z}_{\geq 0}.$$

Assigning a query of type $j \in J$ to resource i earns a finite reward

$$r_{i,j} \geq 0.$$

Queries may always be rejected.

For every $j \in J$, let D_j be a nonnegative, integer-valued random variable. We assume that $D_j < \infty$ almost surely and that the family $(D_j)_{j \in J}$ is mutually independent. Finite support is permitted and is all that will be needed for the upper bound. The lower bound also applies to infinite-support distributions, even when $\mathbb{E}[D_j] = \infty$.

Conditional on $D = (D_j)_{j \in J} = d$, the $\sum_j d_j$ individual queries are placed in a uniformly random permutation. An online policy learns the type of each query when it arrives and must immediately make an irrevocable allocation decision. It is not given d or $\sum_j d_j$ before making its decisions.

For a realization d , let $\text{OFF}(d)$ denote the maximum reward obtainable by a policy that knows all realized demands before allocating. Arrival order does not affect this benchmark. Define

$$\text{OFF}(I) = \mathbb{E}[\text{OFF}(D)]$$

and let $\text{ALG}_{\mathcal{A}}(I)$ denote the expected reward of an online policy $\mathcal{A}$, where the expectation includes demands, arrival order, and the policy's internal randomness.

Because the literal quotient is undefined when $\text{OFF}(I) = 0$, instances with zero benchmark are omitted from the infimum. Equivalently, a factor c means that

$$\text{ALG}_{\mathcal{A}}(I) \geq c \text{OFF}(I)$$

for every instance, including zero-benchmark instances; on such instances this inequality is automatic.

Theorem 1 (Exact competitive ratio). *For independent demands in uniformly random order,*

$$\sup_{\mathcal{A}} \inf_{\substack{I \in \text{Indep} \cap \text{Rand} \\ \text{OFF}(I) > 0}} \frac{\text{ALG}_{\mathcal{A}}(I)}{\text{OFF}(I)} = \frac{1}{2}.$$

The lower bound remains valid under an arbitrary interleaving of the realized queries.

4 A universal one-half policy

4.1 Cloning resources and taking offline marginals

Replace each resource i by k_i distinguishable unit-capacity slots. Let U be the resulting set of slots and put

$$N = |U| = \sum_{i \in R} k_i.$$

If slot u is a clone of resource i , define

$$r_{u,j} = r_{i,j}.$$

An assignment respecting the original capacities can be represented by assigning each accepted query to a distinct corresponding slot, and every slot assignment maps back to a feasible assignment in the original instance. Thus cloning preserves both online feasibility and the pointwise offline value.

If $N = 0$, then $\text{OFF}(I) = 0$, and the desired lower bound is immediate. Henceforth suppose $N \geq 1$.

For every realization $D = d$, fix a deterministically tie-broken offline-optimal matching of queries to slots. Define

$$X_{u,j}(d) = \mathbf{1}\{\text{the fixed offline optimum assigns slot } u \text{ to type } j\}$$

and

$$x_{u,j} = \mathbb{E}[X_{u,j}(D)].$$

For every slot u , pointwise feasibility gives

$$\sum_{j \in J} X_{u,j}(d) \leq 1,$$

and therefore

$$\sum_{j \in J} x_{u,j} \leq 1. \quad (1)$$

For every type j and every subset $S \subseteq U$,

$$\sum_{u \in S} X_{u,j}(d) \leq \min(d_j, |S|).$$

Taking expectations yields

$$x_j(S) := \sum_{u \in S} x_{u,j} \leq \mathbb{E}[\min(D_j, |S|)]. \quad (2)$$

Finally, because the chosen matching is optimal pointwise,

$$\sum_{u \in U} \sum_{j \in J} r_{u,j} x_{u,j} = \mathbb{E}[\text{OFF}(D)] = \text{OFF}(I). \quad (3)$$

4.2 Decomposition into ordered routing lists

We first record the precise finite-dimensional separation statement used in the decomposition.

Lemma 1 (Finite convex-hull criterion). *Let V be a nonempty finite subset of $\mathbb{R}^n$. For $x \in \mathbb{R}^n$,*

$$x \in \text{conv}(V) \iff c \cdot x \leq \max_{v \in V} c \cdot v \text{ for every } c \in \mathbb{R}^n.$$

Proof. The forward implication follows immediately from linearity.

For the converse, let $C = \text{conv}(V)$ and suppose $x \notin C$. Since V is finite, C is the continuous image of a compact simplex and is therefore nonempty and compact. Choose $y \in C$ minimizing $\|x - y\|$. For every $z \in C$ and $t \in [0, 1]$, convexity gives

$$y + t(z - y) \in C.$$

The function

$$g(t) = \|x - y - t(z - y)\|^2$$

has a minimum over $[0, 1]$ at $t = 0$, so its right derivative there is nonnegative:

$$g'(0) = -2(x - y) \cdot (z - y) \geq 0.$$

With $c = x - y$, this says

$$c \cdot z \leq c \cdot y \quad (z \in C).$$

On the other hand,

$$c \cdot x = c \cdot y + \|x - y\|^2 > c \cdot y.$$

Thus

$$c \cdot x > \max_{v \in V} c \cdot v,$$

contradicting the asserted support inequalities. Hence $x \in C$. □

Fix a type j . For $1 \leq \ell \leq N$, set

$$q_{j,\ell} = \Pr(D_j \geq \ell).$$

These tail probabilities satisfy

$$1 \geq q_{j,1} \geq q_{j,2} \geq \cdots \geq q_{j,N} \geq 0.$$

For $0 \leq s \leq N$, define

$$Q_{j,s} = \sum_{\ell=1}^s q_{j,\ell}, \quad Q_{j,0} = 0.$$

The tail-sum identity for a nonnegative integer-valued random variable gives

$$Q_{j,s} = \mathbb{E}[\min(D_j, s)]. \quad (4)$$

Let $\mathcal{L}$ be the family of all ordered lists

$$L = (u_1, \dots, u_s)$$

of pairwise distinct elements of U , where $0 \leq s \leq N$. In particular, $\mathcal{L}$ contains the empty list $L = \emptyset$. For $L = (u_1, \dots, u_s)$, define $z^{j,L} \in \mathbb{R}^U$ by

$$z_{u_\ell}^{j,L} = q_{j,\ell} \quad (1 \leq \ell \leq s), \quad z_u^{j,L} = 0 \quad (u \notin L).$$

For the empty list, $z^{j,\emptyset} = 0$.

Lemma 2 (Ordered-list decomposition). *For fixed j , the polytope*

$$\mathcal{P}_j = \left\{ x \in \mathbb{R}_{\geq 0}^U : \sum_{u \in S} x_u \leq Q_{j,|S|} \text{ for every } S \subseteq U \right\}$$

is exactly

$$\mathcal{P}_j = \text{conv}\{z^{j,L} : L \in \mathcal{L}\}.$$

Proof. First consider a list $L = (u_1, \dots, u_s)$. If $S \subseteq U$ has cardinality a , then the coordinates of $z^{j,L}$ indexed by S consist of at most a members of the nonincreasing sequence

$$q_{j,1}, \dots, q_{j,s}.$$

Consequently,

$$\sum_{u \in S} z_u^{j,L} \leq \sum_{\ell=1}^{\min(a,s)} q_{j,\ell} \leq Q_{j,a}.$$

Thus every list vector belongs to $\mathcal{P}_j$, and convexity gives

$$\text{conv}\{z^{j,L} : L \in \mathcal{L}\} \subseteq \mathcal{P}_j.$$

Conversely, take $x \in \mathcal{P}_j$ and an arbitrary coefficient vector $c \in \mathbb{R}^U$. If c has no positive coordinate, then $x \geq 0$ implies

$$c \cdot x \leq 0 = c \cdot z^{j,\emptyset} \leq \max_{L \in \mathcal{L}} c \cdot z^{j,L}.$$

Suppose now that c has $s \geq 1$ positive coordinates. Label them so that

$$c_{u_1} \geq c_{u_2} \geq \cdots \geq c_{u_s} > 0$$

and put $c_{u_{s+1}} = 0$. Since all coordinates of x are nonnegative, the nonpositive coordinates of c may only decrease $c \cdot x$. Therefore

$$c \cdot x \leq \sum_{\ell=1}^s c_{u_\ell} x_{u_\ell}.$$

Let

$$S_\ell = \{u_1, \dots, u_\ell\}.$$

Summation by parts and the defining inequalities of $\mathcal{P}_j$ give

$$\begin{aligned} c \cdot x &\leq \sum_{\ell=1}^s (c_{u_\ell} - c_{u_{\ell+1}}) x(S_\ell) \\ &\leq \sum_{\ell=1}^s (c_{u_\ell} - c_{u_{\ell+1}}) Q_{j,\ell} \\ &= \sum_{\ell=1}^s c_{u_\ell} q_{j,\ell}. \end{aligned}$$

The last expression is the maximum of $c \cdot z^{j,L}$ over all lists L . Indeed, deleting nonpositive coordinates from a list cannot decrease its value: their own contributions are nonpositive, while each retained positive coordinate can only move to an earlier position and hence receive a weakly larger $q_{j,\ell}$. Among positive coordinates, an adjacent interchange shows that placing the larger coefficient earlier cannot decrease the value because the sequence $(q_{j,\ell})$ is nonincreasing. Finally, adding any omitted positive coordinate at the end cannot decrease the value. Thus the maximizing list consists of all positive coordinates ordered as $u_1, \dots, u_s$, and its value is exactly

$$\sum_{\ell=1}^s c_{u_\ell} q_{j,\ell}.$$

We have therefore proved, for every $c \in \mathbb{R}^U$,

$$c \cdot x \leq \max_{L \in \mathcal{L}} c \cdot z^{j,L}.$$

The hypotheses of Lemma 1 hold here: U is finite, so the ambient dimension is $N < \infty$; the family of ordered lists is finite; and it is nonempty because it includes the empty list. Applying that lemma proves

$$x \in \text{conv}\{z^{j,L} : L \in \mathcal{L}\}.$$

□

The empty-list convention is essential. It represents the zero vector and ensures that the support maximum is zero when all coordinates of c are nonpositive.

By (2) and (4), the vector $x_{\cdot,j}$ belongs to $\mathcal{P}_j$. Lemma 2 therefore gives coefficients

$$\lambda_{j,L} \geq 0, \quad \sum_{L \in \mathcal{L}} \lambda_{j,L} = 1,$$

such that

$$x_{\cdot,j} = \sum_{L \in \mathcal{L}} \lambda_{j,L} z^{j,L}. \quad (5)$$

Independently for every type j , sample an ordered list

$$L_j = (u_{j,1}, \dots, u_{j,s_j})$$

according to $(\lambda_{j,L})_{L \in \mathcal{L}}$. These lists are sampled independently of all demands and of the arrival order. On the ℓ -th arrival of type j , offer that query to $u_{j,\ell}$ if $\ell \leq s_j$; if $\ell > s_j$, reject it.

For a fixed slot u , let $Z_{u,j}$ be the event that some type- j query is offered to u . Because every list has distinct slots, there is at most one such query. If u has position ℓ in L_j , then $Z_{u,j}$ occurs exactly when $D_j \geq \ell$. Consequently,

$$\begin{aligned} \Pr(Z_{u,j}) &= \sum_{\substack{L=(u_1, \dots, u_s) \in \mathcal{L} \\ u=u_\ell \text{ for some } \ell}} \lambda_{j,L} \Pr(D_j \geq \ell) \\ &= \sum_{L \in \mathcal{L}} \lambda_{j,L} z_u^{j,L} = x_{u,j}. \end{aligned} \quad (6)$$

Moreover, for fixed u , the events $(Z_{u,j})_{j \in J}$ are mutually independent. Indeed, $Z_{u,j}$ is determined only by the pair (L_j, D_j) , and these pairs are mutually independent across types.

4.3 An order-independent contention lemma

Lemma 3 (Independent contention). *Let J_0 be a finite set of candidates. Candidate $j \in J_0$ is active independently with probability p_j , has value $v_j \geq 0$, and*

$$P := \sum_{j \in J_0} p_j \leq 1.$$

The active candidates may be presented in an arbitrary order, even an order depending on the complete active set. There is an online policy that accepts at most one candidate and has expected value at least

$$\frac{1}{2} \sum_{j \in J_0} p_j v_j.$$

Proof. Let

$$V = \sum_{j \in J_0} p_j v_j, \quad \tau = \frac{V}{2},$$

and define

$$H = \{j \in J_0 : v_j \geq \tau\}.$$

The policy accepts the first presented active candidate belonging to H and rejects every candidate outside H . This policy is online and does not use any information about future presentations.

Let

$$\theta = \prod_{j \in H} (1 - p_j).$$

The policy earns at least τ whenever at least one member of H is active. In addition, when j is the unique active member of H , its reward is v_j , so beyond the baseline τ it earns the surplus $v_j - \tau$. Hence

$$\mathbb{E}[\text{reward}] \geq \tau(1 - \theta) + \sum_{j \in H} (v_j - \tau)p_j \prod_{\ell \in H \setminus \{j\}} (1 - p_\ell).$$

For every $j \in H$,

$$\prod_{\ell \in H \setminus \{j\}} (1 - p_\ell) \geq \prod_{\ell \in H} (1 - p_\ell) = \theta.$$

This inequality remains valid when some p_j is 0 or 1; no division by $1 - p_j$ is used. Since $v_j - \tau \geq 0$ on H ,

$$\mathbb{E}[\text{reward}] \geq \tau(1 - \theta) + \theta \sum_{j \in H} (v_j - \tau)p_j.$$

For $j \notin H$, $v_j - \tau < 0$. Omitting these negative terms gives

$$\begin{aligned} \sum_{j \in H} (v_j - \tau)p_j &\geq \sum_{j \in J_0} (v_j - \tau)p_j \\ &= V - \tau P \\ &\geq V - \tau \\ &= \tau, \end{aligned}$$

where $P \leq 1$, $\tau \geq 0$, and $V = 2\tau$ were used. Therefore

$$\mathbb{E}[\text{reward}] \geq \tau(1 - \theta) + \theta\tau = \tau = \frac{V}{2}.$$

The case $V = 0$ is included: then $\tau = 0$, and the claimed lower bound is zero. $\square$

4.4 Combining routing and contention

For every slot u , define

$$p_{u,j} = x_{u,j}, \quad v_{u,j} = r_{u,j}.$$

Equation (1) gives

$$\sum_{j \in J} p_{u,j} \leq 1.$$

By (6), type j offers a candidate to u with probability $p_{u,j}$, and these activation events are independent across j .

At slot u , run the policy of Lemma 3. Equivalently, define

$$V_u = \sum_{j \in J} x_{u,j} r_{u,j}, \quad \tau_u = \frac{V_u}{2},$$

and, while u is unoccupied, accept the first query offered to u whose reward is at least τ_u . All other queries offered to u are rejected.

Each arriving query is offered to at most one slot, and every slot accepts at most one query. Hence the resulting slotwise decisions form a legal online allocation. The presentation order at a slot may depend on all realized demands, all sampled lists, and the global arrival order. Lemma 3 nevertheless applies because its guarantee is valid for an arbitrary order, even one determined by the complete active set.

For each u ,

$$\mathbb{E}[\text{reward earned at } u] \geq \frac{1}{2} \sum_{j \in J} x_{u,j} r_{u,j}.$$

Summing over slots and using (3),

$$\begin{aligned} \text{ALG}(I) &\geq \frac{1}{2} \sum_{u \in U} \sum_{j \in J} r_{u,j} x_{u,j} \\ &= \frac{1}{2} \text{OFF}(I). \end{aligned} \tag{7}$$

Only the tail probabilities $q_{j,1}, \dots, q_{j,N}$ enter this construction. Thus infinite support and even infinite mean of D_j cause no difficulty: capacities bound both online and offline rewards. Zero demands, zero capacities, zero rewards, activation probabilities in $\{0,1\}$, and thresholds with $V_u = 0$ are all covered by the same argument. Most importantly, no property of the arrival order was used, so (7) holds under arbitrary interleaving.

5 A matching upper bound

5.1 The hard family

Fix an integer $M \geq 1$ and a real number $h \geq 2$. Consider one resource of capacity one and two query types:

- a low type L with reward 1;
- a high type G with reward h .

Let

$$H_M = \sum_{d=1}^M \frac{1}{d}.$$

The demands are independent and distributed as

$$\Pr(D_G = 1) = \frac{1}{h}, \quad \Pr(D_G = 0) = 1 - \frac{1}{h},$$

and

$$\Pr(D_L = d) = \frac{1}{H_M d}, \quad d = 1, \dots, M. \tag{8}$$

The low-demand distribution is normalized because

$$\sum_{d=1}^M \frac{1}{H_M d} = 1.$$

Both demands have finite support, and conditional on their realization the queries are presented in a uniformly random permutation. Denote this instance by $I_{h,M}$.

Since $D_L \geq 1$ almost surely, the offline optimum receives 1 when $D_G = 0$. When $D_G = 1$, it assigns the high query and receives h . Therefore

$$\text{OFF}(I_{h,M}) = \left(1 - \frac{1}{h}\right) 1 + \frac{1}{h} h = 2 - \frac{1}{h}. \quad (9)$$

5.2 Reduction of arbitrary policies to stopping indices

Consider any randomized, adaptive online policy. It is without loss of generality to accept the high query whenever the resource is still available. Indeed, there is at most one high query, accepting it yields h , and after rejecting it every possible future reward is at most 1. Replacing rejection of an available high query by acceptance therefore improves the reward pathwise.

Pre-sample the policy's entire random tape independently of the demands and the arrival order, and temporarily fix that tape. Before the high query appears, every observed query is of type L . Because the policy is not told the realized total demand, its history at such a point consists only of a number of preceding low arrivals, their rejections, and the fixed internal random tape.

On a hypothetical infinite history containing only low arrivals, let

$$T \in \mathbb{N} \cup \{\infty\}$$

be the index of the first low query accepted by the resulting deterministic policy; set $T = \infty$ if it never accepts a low query. Thus, until a high query appears, the deterministic policy is characterized by this first-acceptance index. Restoring the random tape makes T a random variable independent of the realized demands and arrival order. Consequently, every randomized history-dependent policy is a mixture of the deterministic policies indexed by T . This is a direct random-tape argument and does not invoke Yao's principle.

Fix first $T = t < \infty$, and define

$$S_t = \Pr(D_L \geq t)$$

and

$$b_t = \mathbb{E} \left[\mathbf{1}_{\{D_L \geq t\}} \frac{t}{D_L + 1} \right].$$

If $D_G = 0$, the policy earns 1 precisely when $D_L \geq t$, so its conditional expected reward contribution is

$$\left(1 - \frac{1}{h}\right) S_t.$$

Suppose $D_G = 1$. If $D_L = d < t$, the t -th low query does not exist, so the high query is eventually encountered while the capacity is available and is accepted for reward h . If $d \geq t$, then among the $d+1$ uniformly permuted queries, the high query has each possible rank relative to the d low queries with probability $1/(d+1)$. It precedes the t -th low query in exactly t of these $d+1$ possible relative positions. Hence

$$\Pr(\text{high precedes the } t\text{-th low} \mid D_L = d, D_G = 1) = \frac{t}{d+1}.$$

On that event the policy earns h ; otherwise it accepts the t -th low query first and earns 1.

It follows that

$$\begin{aligned}\mathbb{E}[\text{ALG}_t] &= \left(1 - \frac{1}{h}\right) S_t + \frac{1}{h} [h(1 - S_t) + S_t + (h - 1)b_t] \\ &= 1 + \left(1 - \frac{1}{h}\right) b_t.\end{aligned}\tag{10}$$

For $t = \infty$, the policy never accepts a low query. It earns h exactly when $D_G = 1$, so its expected reward is 1. Thus (10) remains valid if we define $b_\infty = 0$.

For $1 \leq t \leq M$, distribution (8) gives

$$\begin{aligned}b_t &= \frac{t}{H_M} \sum_{d=t}^M \frac{1}{d(d+1)} \\ &= \frac{t}{H_M} \sum_{d=t}^M \left(\frac{1}{d} - \frac{1}{d+1}\right) \\ &= \frac{t}{H_M} \left(\frac{1}{t} - \frac{1}{M+1}\right) \\ &= \frac{1 - t/(M+1)}{H_M} \leq \frac{1}{H_M}.\end{aligned}\tag{11}$$

For $t > M$, the event $D_L \geq t$ is impossible, and hence $b_t = 0$.

The quantity in (11) is decreasing in t for $1 \leq t \leq M$. Therefore the best deterministic stopping index is $t = 1$, and averaging over a randomized T cannot do better. In particular, accepting the first low query is exactly optimal within this instance family, and every randomized adaptive online policy satisfies

$$\begin{aligned}\text{ALG}(I_{h,M}) &\leq 1 + \left(1 - \frac{1}{h}\right) b_1 \\ &= 1 + \left(1 - \frac{1}{h}\right) \frac{M}{(M+1)H_M} \\ &\leq 1 + \frac{1 - 1/h}{H_M}.\end{aligned}\tag{12}$$

5.3 Limiting ratio

Combining (9) and (12), every online policy satisfies

$$\frac{\text{ALG}(I_{h,M})}{\text{OFF}(I_{h,M})} \leq \frac{1 + (1 - 1/h)/H_M}{2 - 1/h}.\tag{13}$$

The harmonic numbers diverge, since

$$H_M = \sum_{d=1}^M \frac{1}{d} \geq \int_1^{M+1} \frac{dx}{x} = \log(M+1).$$

Choose, for example, $h = M$ and let $M \rightarrow \infty$. The right-hand side of (13) then converges to

$$\frac{1}{2}.$$

Explicitly, for every $\varepsilon > 0$, there exists M_ε such that for every $M \geq M_\varepsilon$, every online policy on $I_{M,M}$ satisfies

$$\frac{\text{ALG}(I_{M,M})}{\text{OFF}(I_{M,M})} \leq \frac{1}{2} + \varepsilon.$$

Thus, for every online algorithm $\mathcal{A}$,

$$\inf_{\substack{I \in \text{Indep} \cap \text{Rand} \\ \text{OFF}(I) > 0}} \frac{\text{ALG}_{\mathcal{A}}(I)}{\text{OFF}(I)} \leq \frac{1}{2}.$$

No claim that a finite instance attains ratio exactly $1/2$ is needed; the sequence of finite instances suffices for the infimum.

Together with the universal lower bound (7), this proves Theorem 1. The lower and upper bounds coincide, so the paper provides a complete solution and leaves no remaining gap in the stated model.

References

- [1] Ali Aouad, Will Ma. *A Nonparametric Framework for Online Stochastic Matching with Correlated Arrivals*. Economics and Computation (EC), 2023. <https://doi.org/10.1145/3580507.3597773>